

Cell Theory & Historical Milestones

Cell: The fundamental structural and functional unit of all living organisms.



Robert Hooke:
Observed cork cells.



Antonie Van Leeuwenhoek:
First to see & describe a LIVE cell.



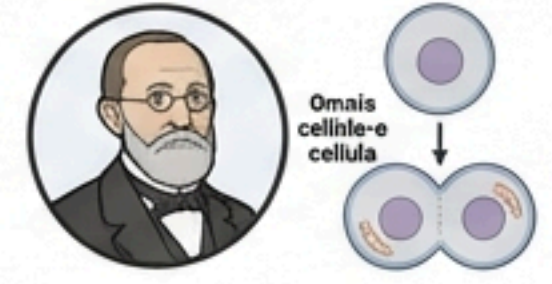
Robert Brown (1831):
Discovered the Nucleus.



Schleiden (1838):
German Botanist.
Plants = aggregates of cells.



Schwann (1839):
German Zoologist.
Animal cells have plasma membrane.

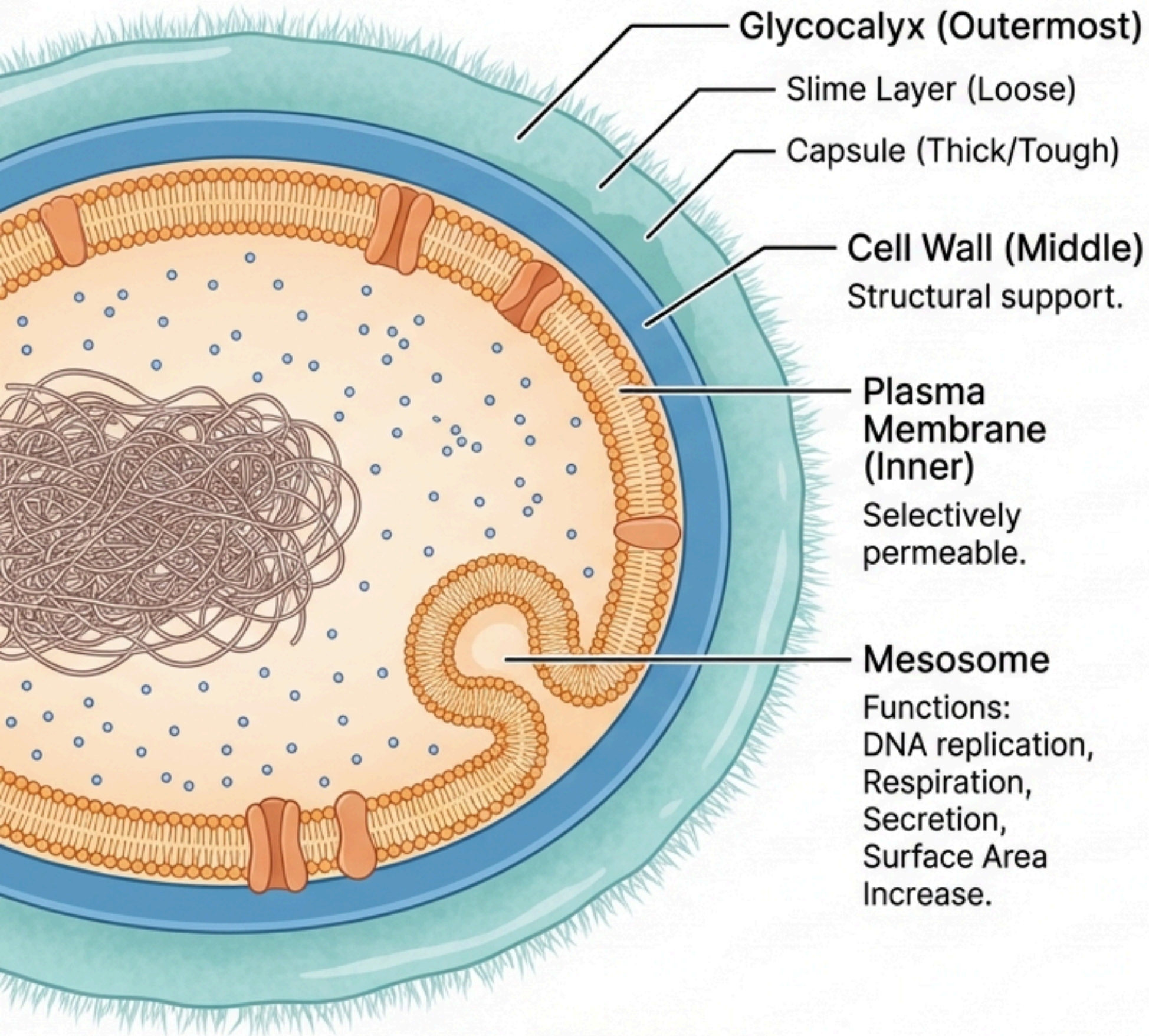


Rudolf Virchow (1855):
Omnis cellula-e cellula
(Cells arise from pre-existing cells).

High Yield
Proposed that Cell Wall is unique to plants.

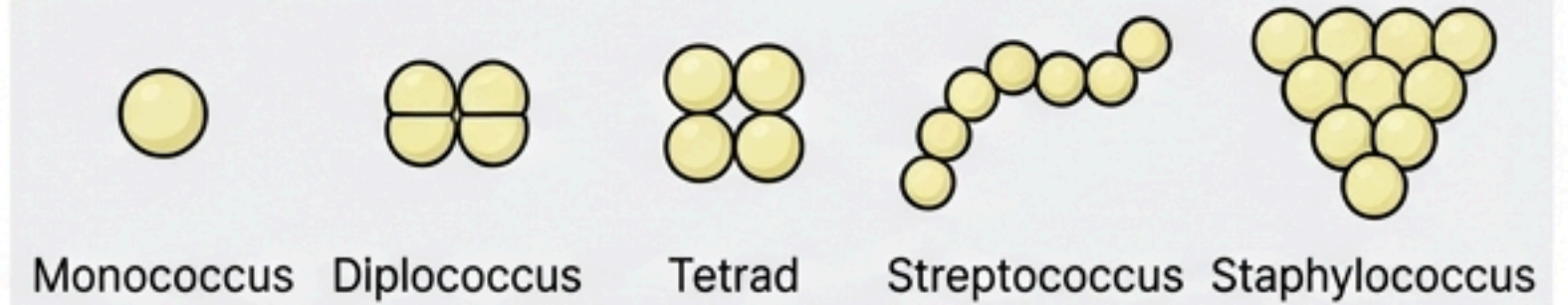
Schleiden & Schwann (Formulation) + Virchow (Modification) = **Modern Cell Theory**
SOHAN KUMAR

Prokaryotic Cells: Envelope & Modifications

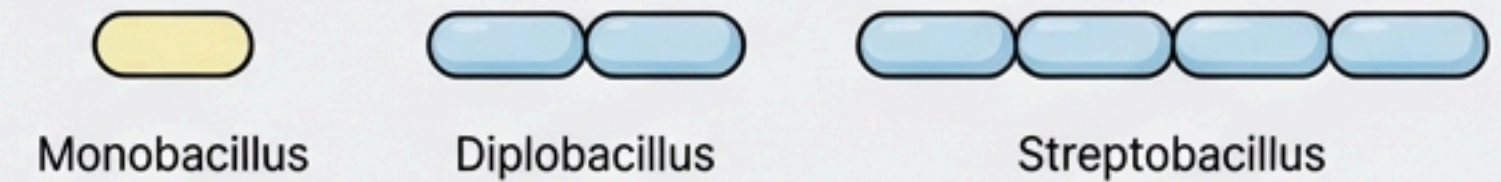


Bacterial Shapes

Cocci



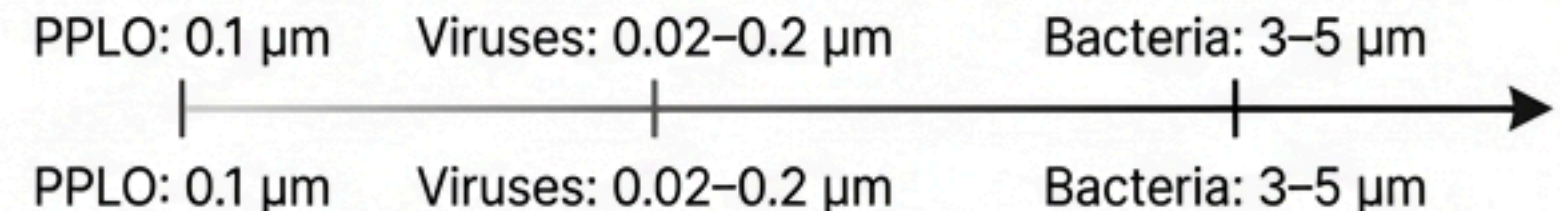
Bacilli



Spirilla

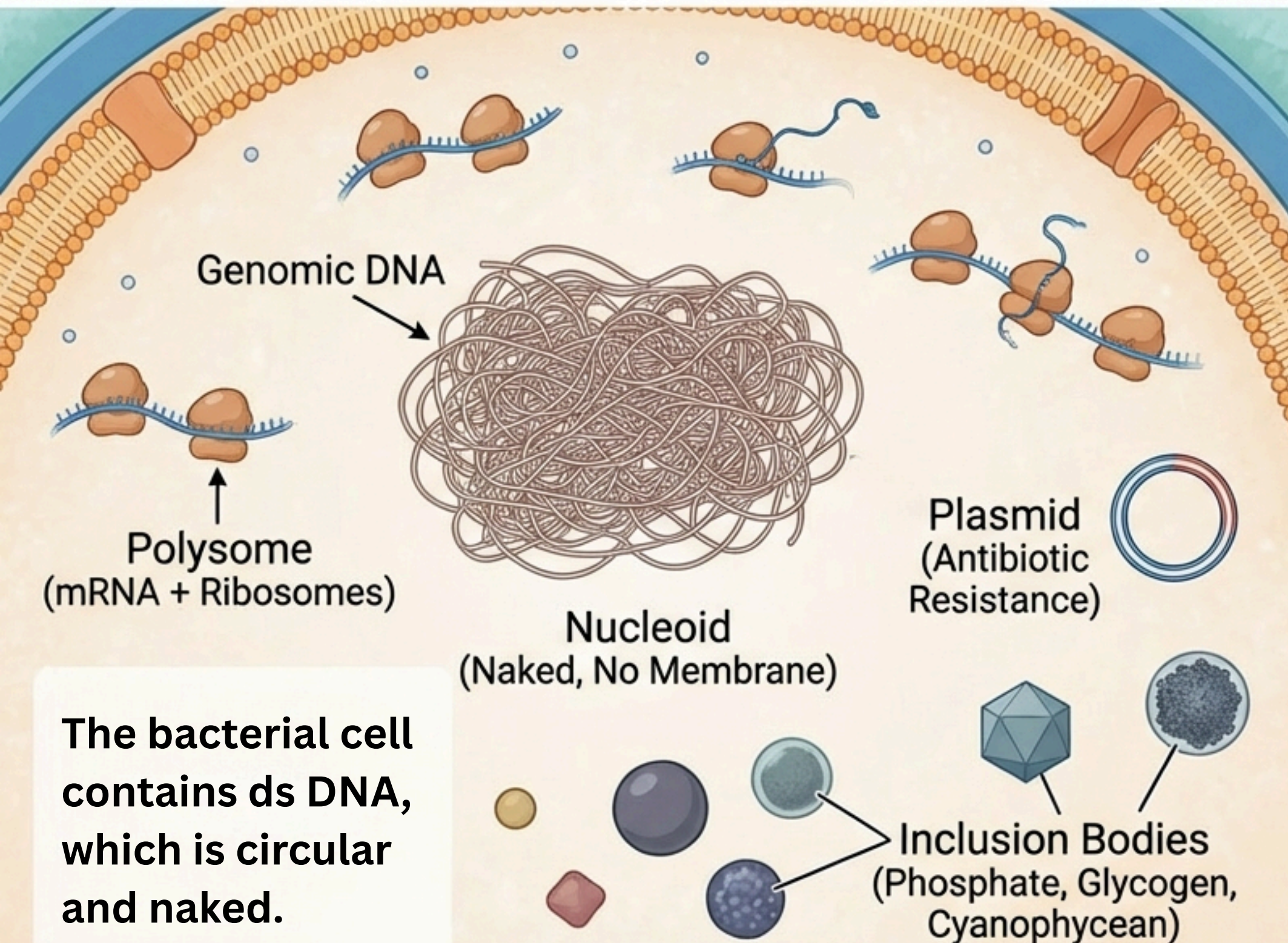


Vibrio



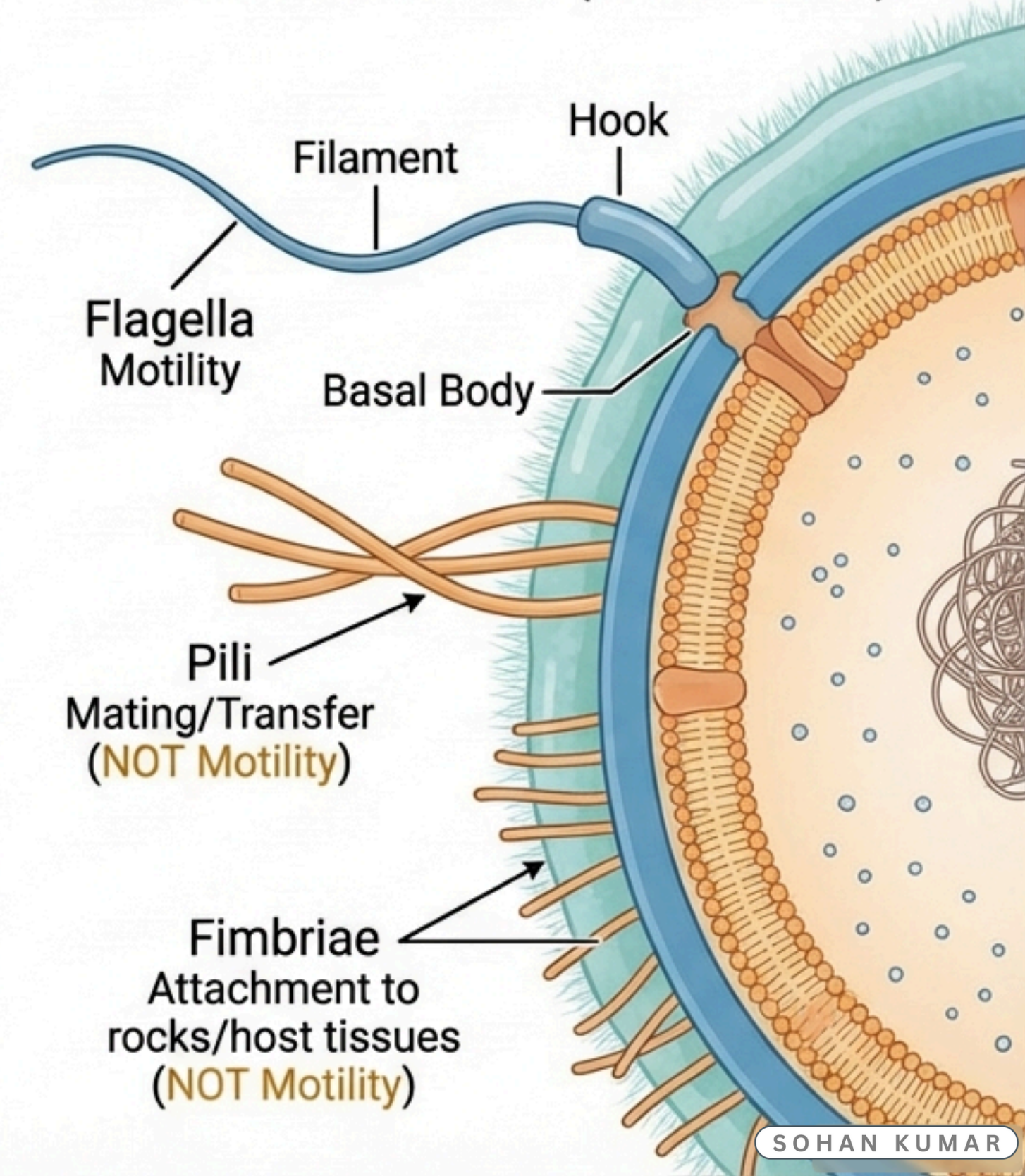
Prokaryotic Cytoplasm: Genetic Material & Surface

Internal Machinery

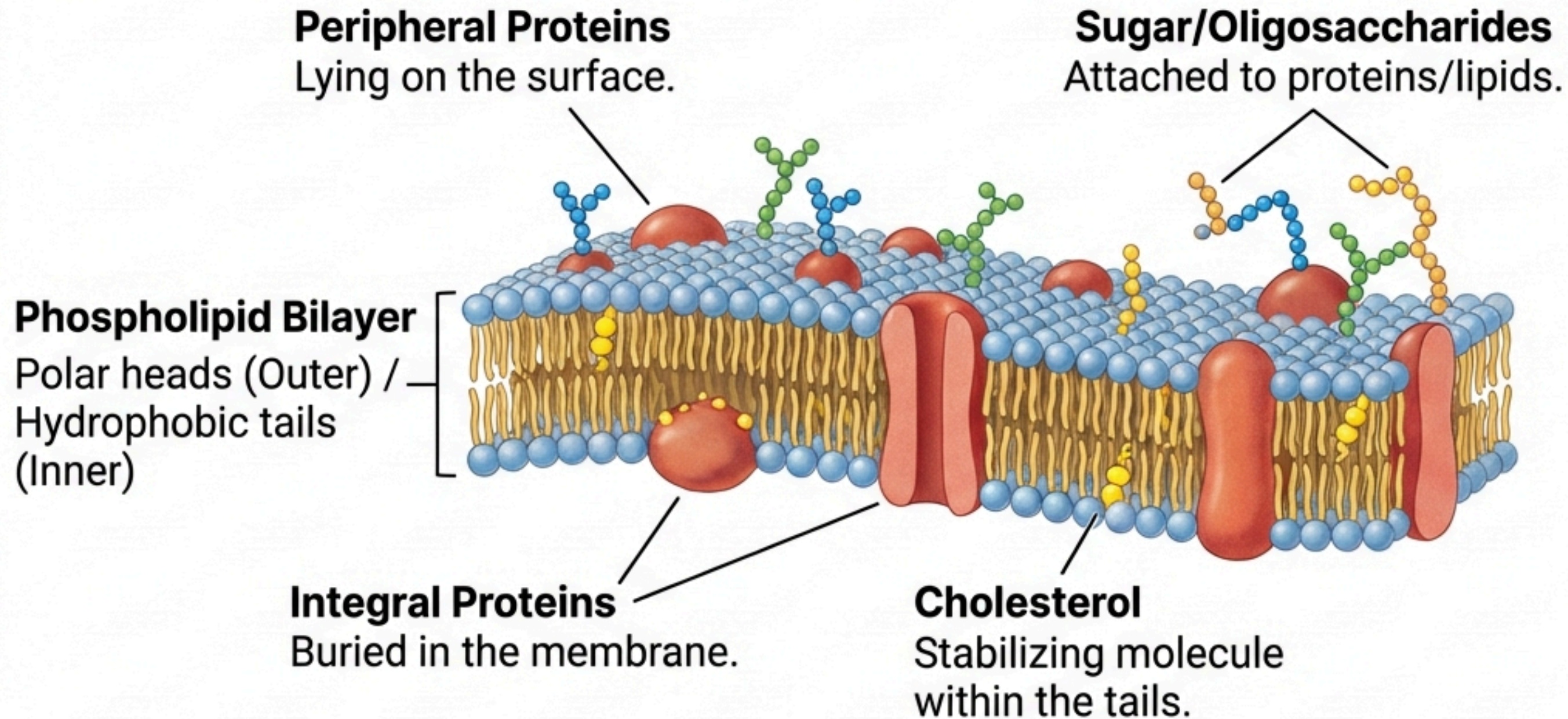


The bacterial cell contains ds DNA, which is circular and naked.

Surface Structures (NEET TRAP)



Cell Membrane: Fluid Mosaic Model (Singer & Nicolson, 1972)



Quasi-fluid Nature

Enables lateral movement of proteins.

Essential for:

- Cell growth
- Secretion
- Endocytosis
- Division.

Transport Mechanisms

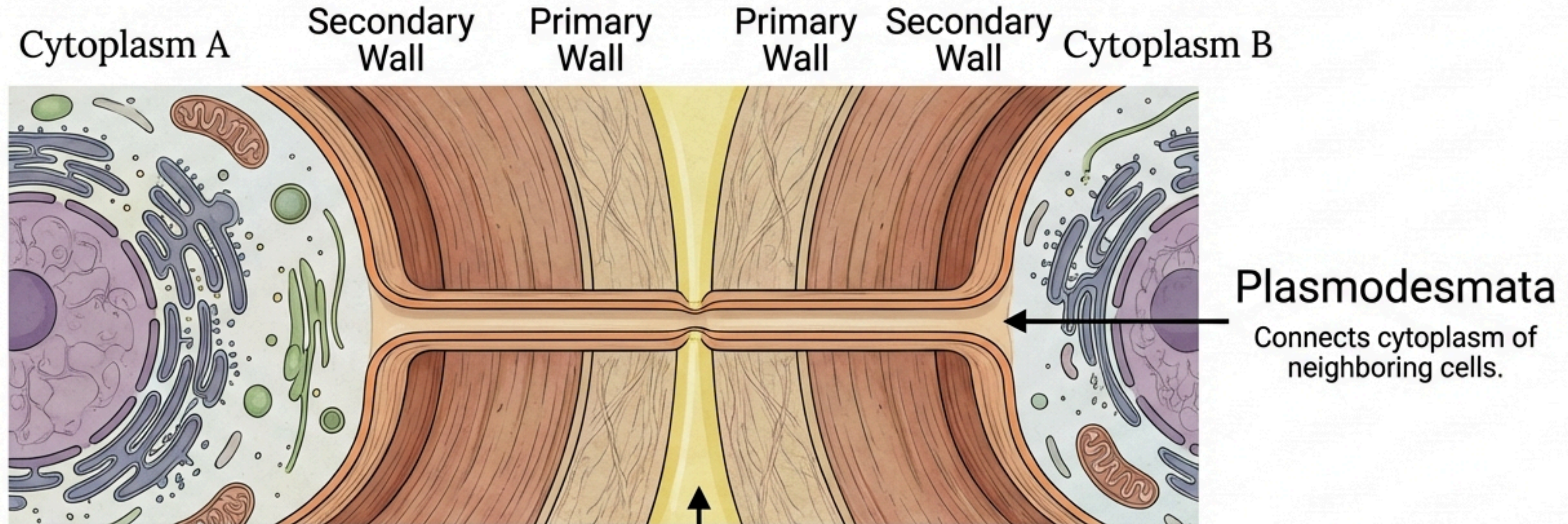
Passive

Diffusion (Neutral solutes) & Osmosis (Water).

Active

Against gradient, requires ATP (e.g., Na^+/K^+ Pump).

Cell Wall & Intercellular Connections



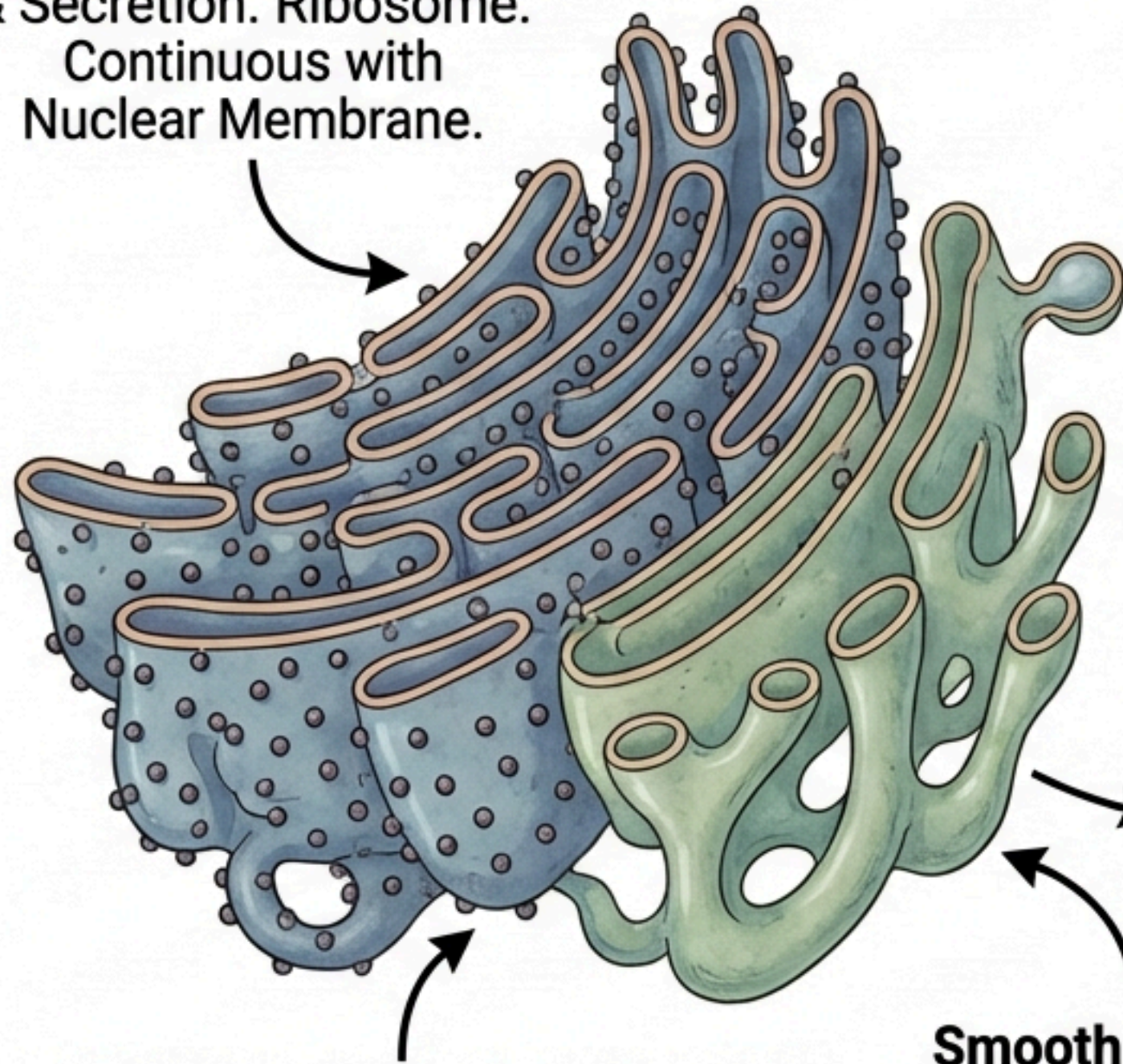
Secondary wall forms on the INNER side (towards membrane) as cell matures.

Middle Lamella
Calcium Pectate
(Glues cells together)

Cell Wall Composition	
Algae →	Cellulose, Galactans, Mannans, CaCO ₃
Plants →	Cellulose, Hemicellulose, Pectins, Proteins

Endomembrane System I: ER & Golgi Complex

RER: Protein Synthesis & Secretion. Ribosome. Continuous with Nuclear Membrane.



RER: Protein Synthesis & Secretion. Continuous with Nuclear Membrane.

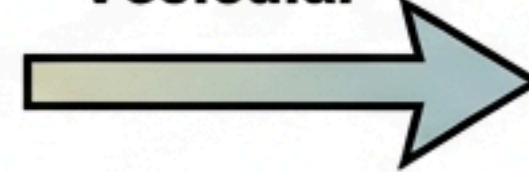
Smooth ER (SER)
SER: Lipid & Steroidal Hormone Synthesis.

Golgi Apparatus

Cis Face
(Forming/Convex)

Trans Face
(Maturing/Concave)

Vesicular



Vesicular Transport

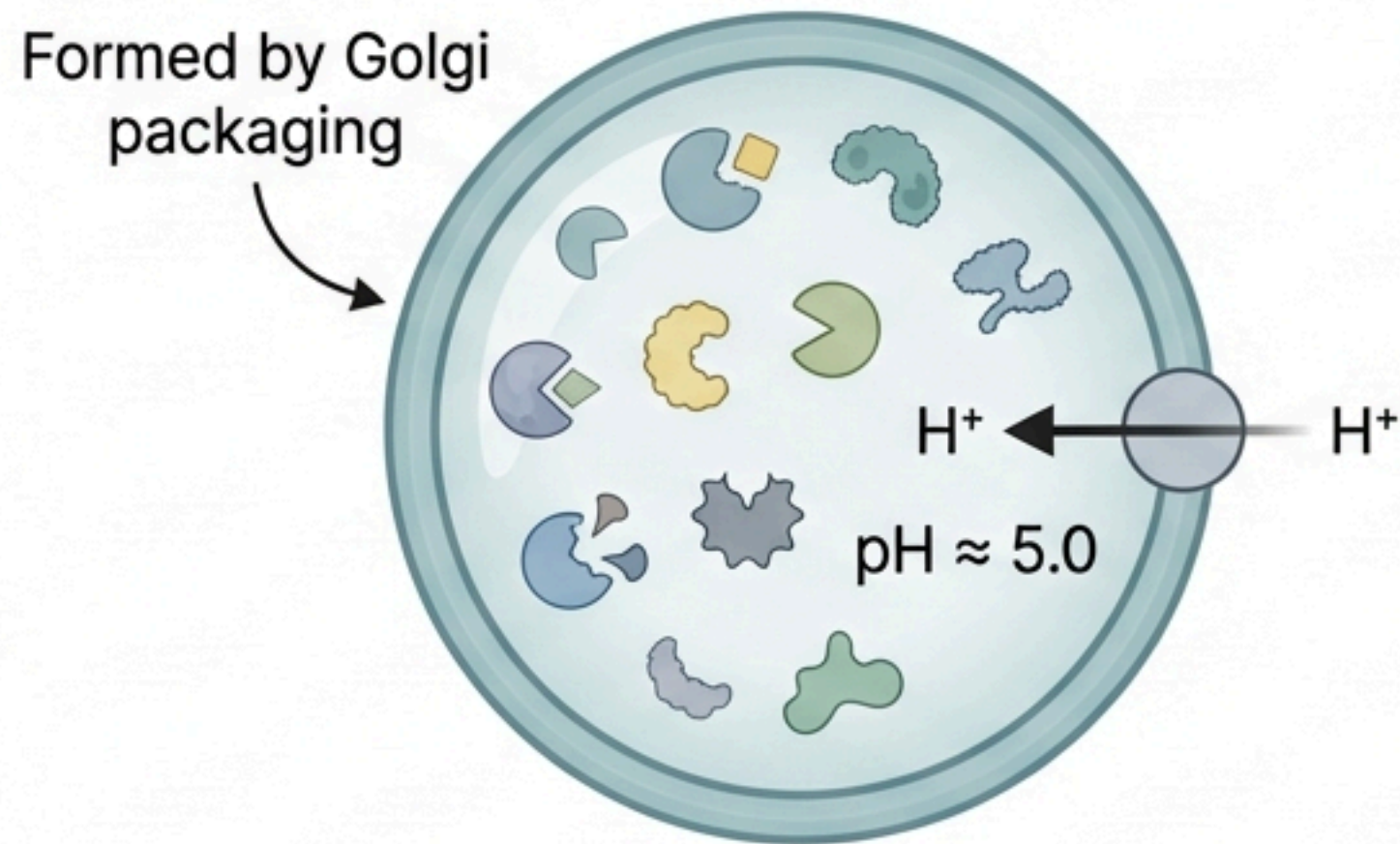
Cis Face
(Forming/Convex)

Trans Face
(Maturing/Concave)

Site of **Glycosylation**
(Formation of Glycoproteins & Glycolipids)

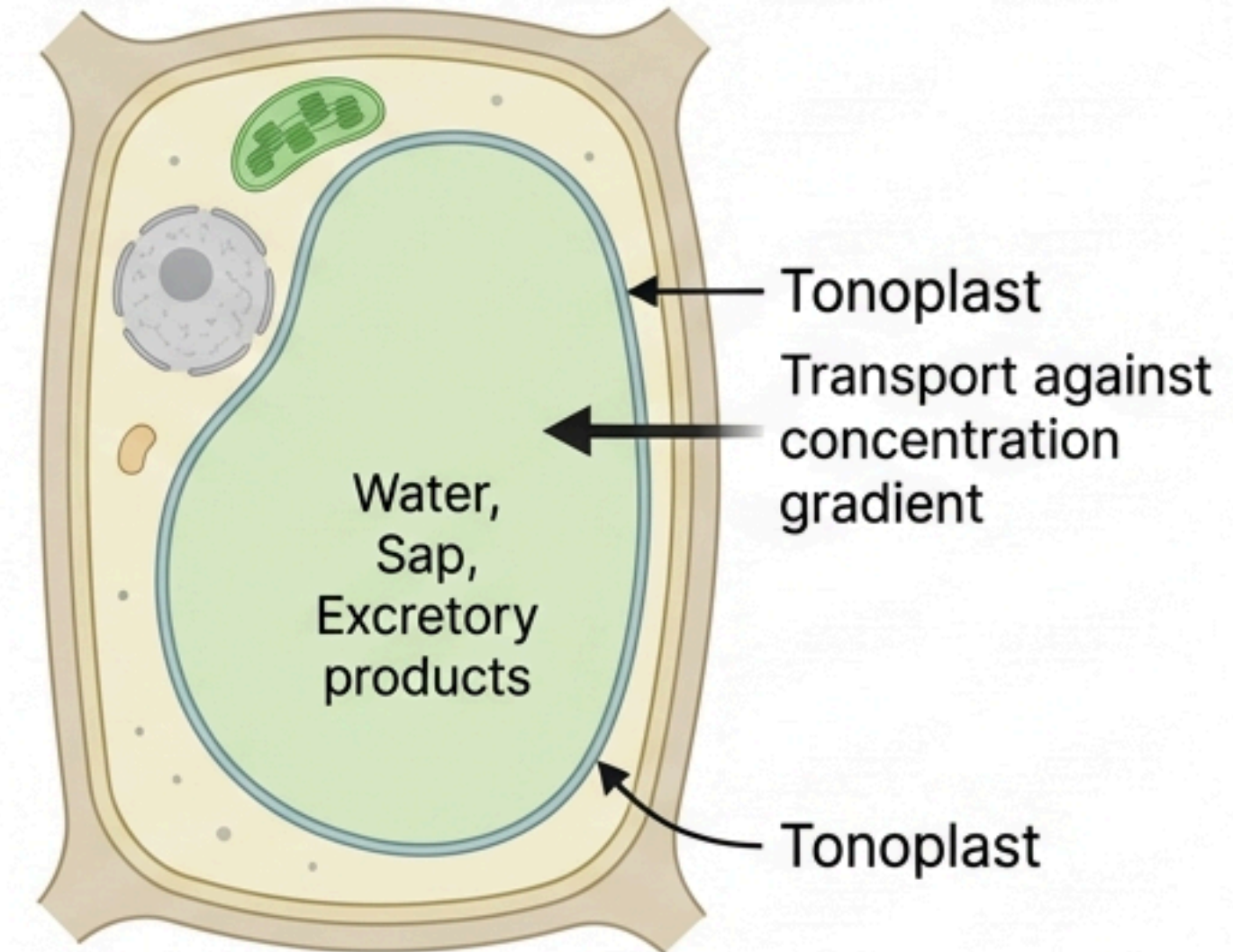
Endomembrane System II: Lysosomes & Vacuoles

Lysosomes (Suicidal Bags)



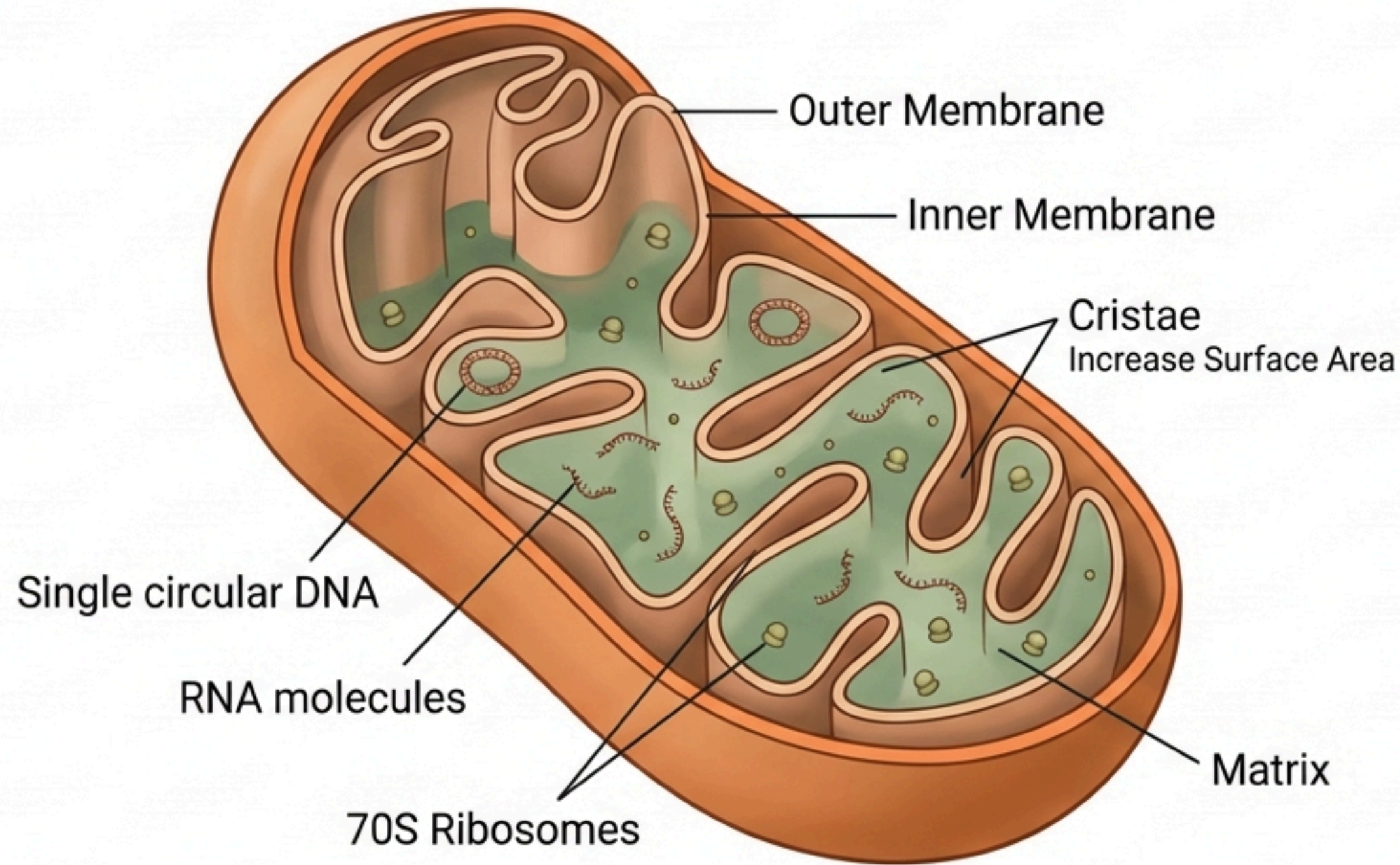
- Rich in Hydrolytic Enzymes (Lipases, Proteases, Carbohydrases).
- Active at ACIDIC pH.

Vacuoles



- Contractile Vacuule (Amoeba) → Osmoregulation.
- Food Vacuole (Protists) → Engulfing food.

Mitochondria: The Powerhouse



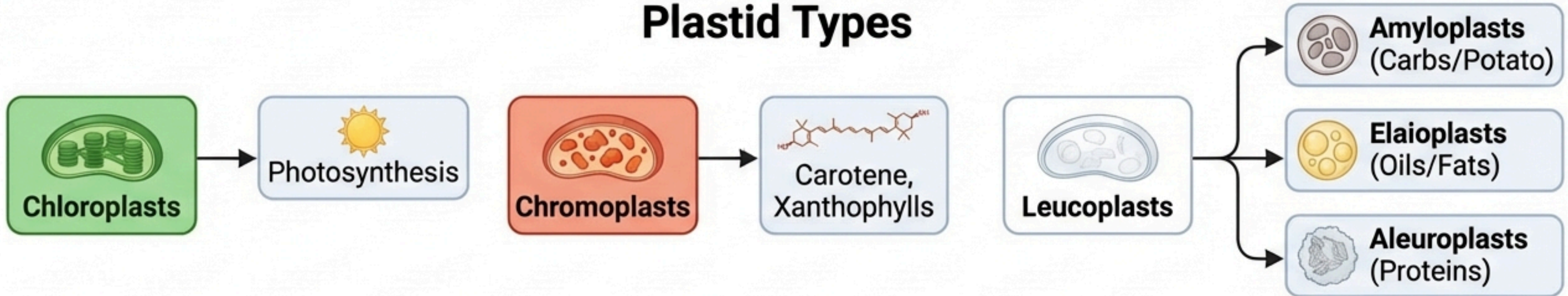
Function: Aerobic Respiration (ATP Production)

Division: Fission

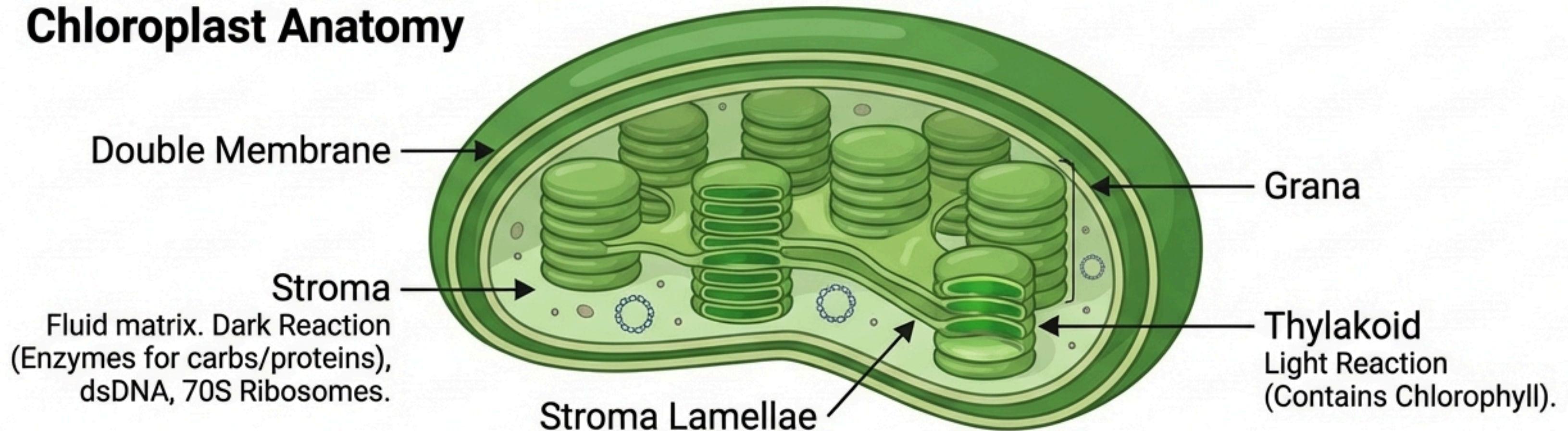
Semi-autonomous:
Has own DNA and Ribosomes

Plastids: Classification & Chloroplast Structure

Plastid Types

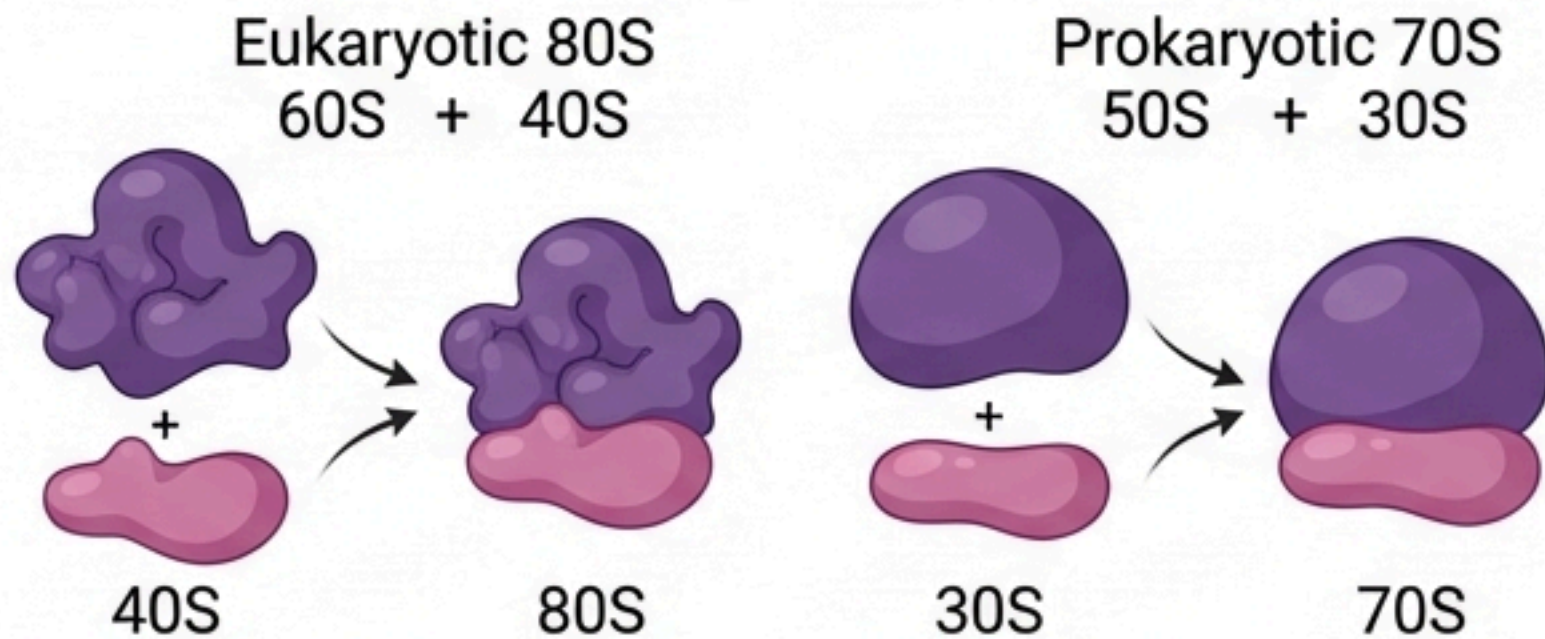


Chloroplast Anatomy



Ribosomes & Cytoskeleton

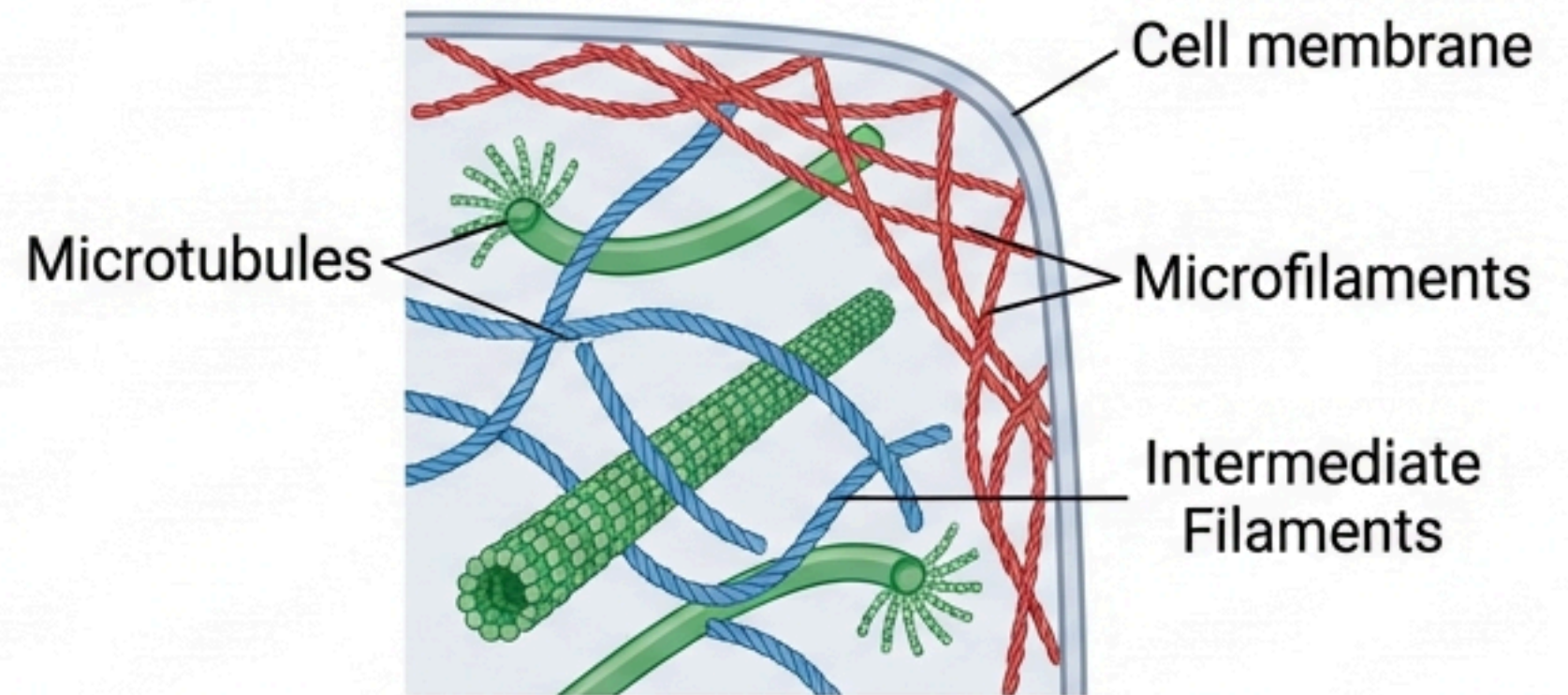
Ribosomes (George Palade, 1953)



Composed of RNA + Protein.
Non-membrane bound.

‘S’ = Svedberg Unit (Sedimentation coefficient - density/size).

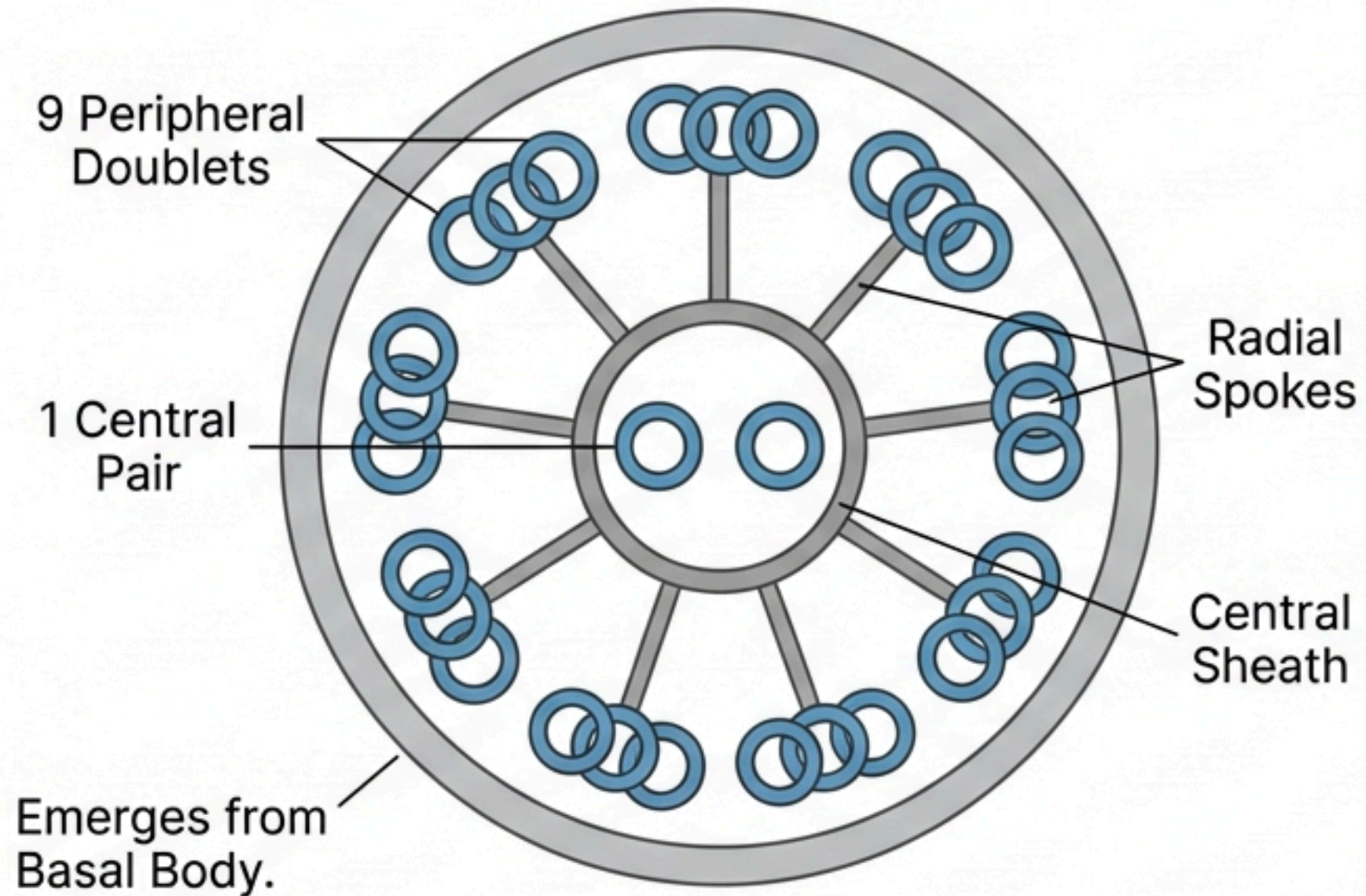
Cytoskeleton



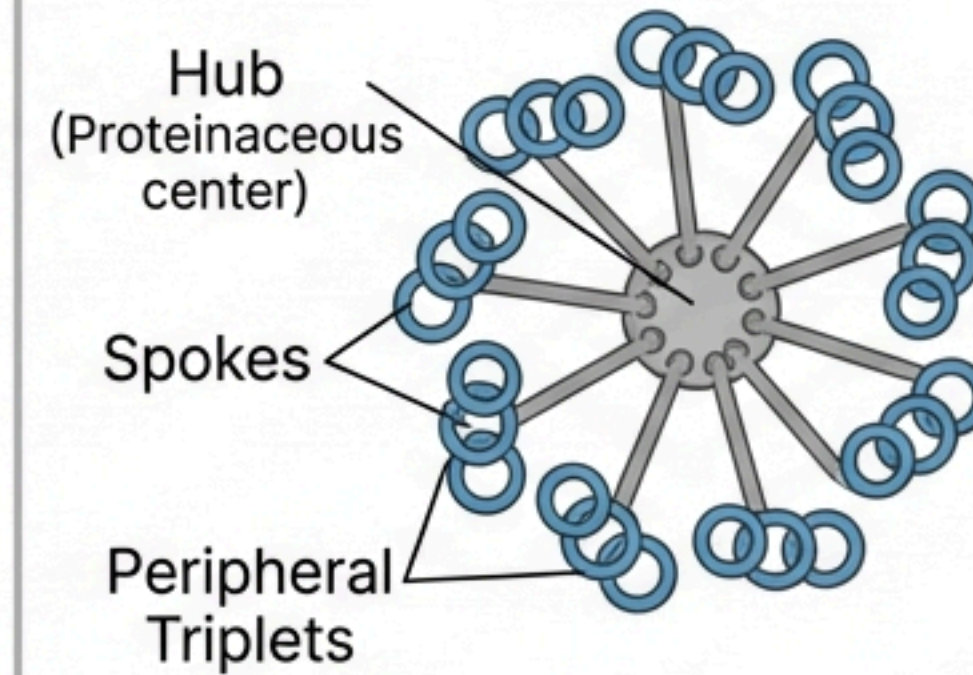
Mechanical support,
Motility,
Maintenance of shape.

Cilia, Flagella & Centrosomes

The Axoneme
(9+2 Arrangement)



Centriole
(9+0 Arrangement)

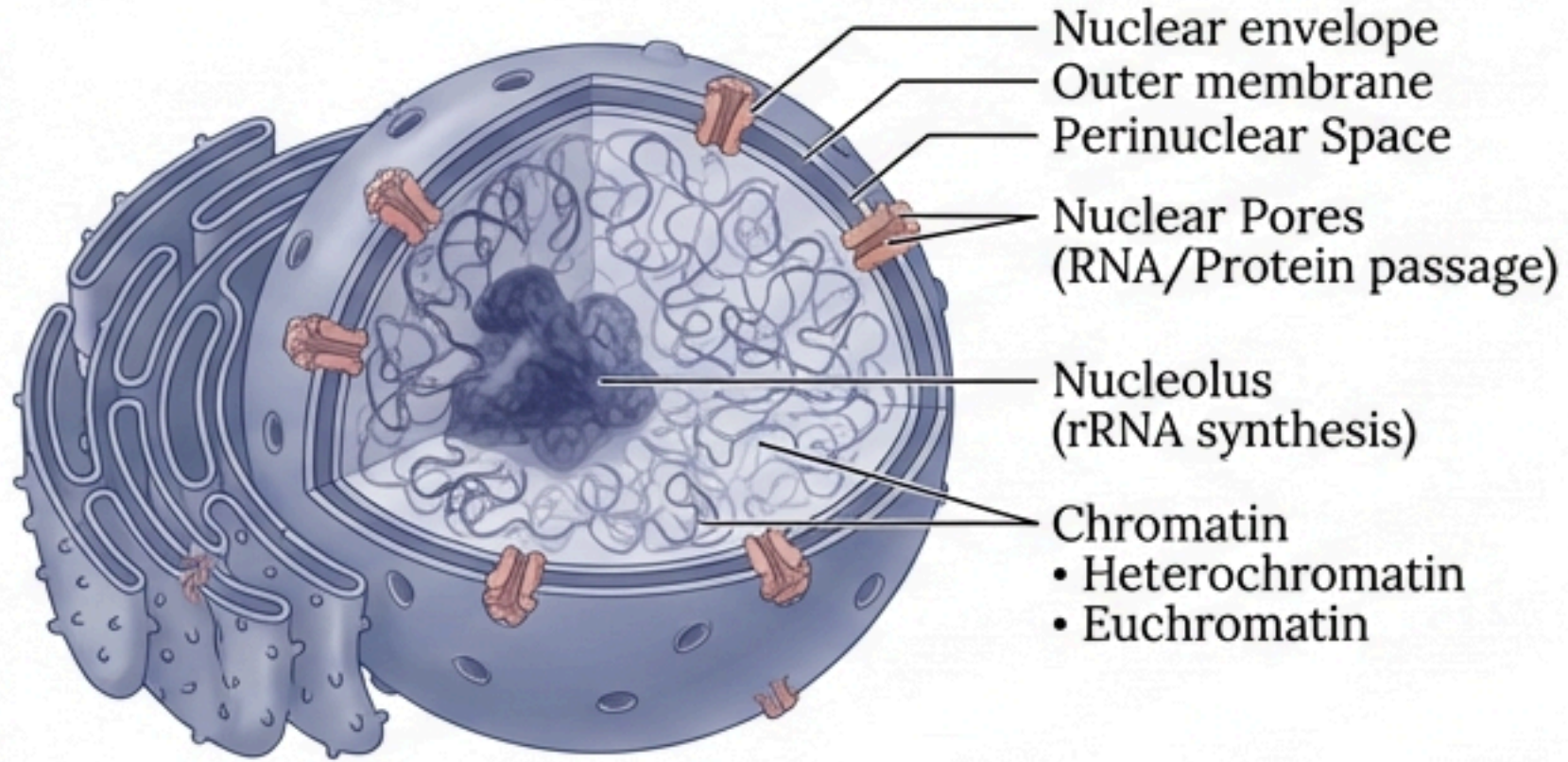


Centrosome
= 2 Centrioles
perpendicular
to each other.
Forms Spindle
Fibres.

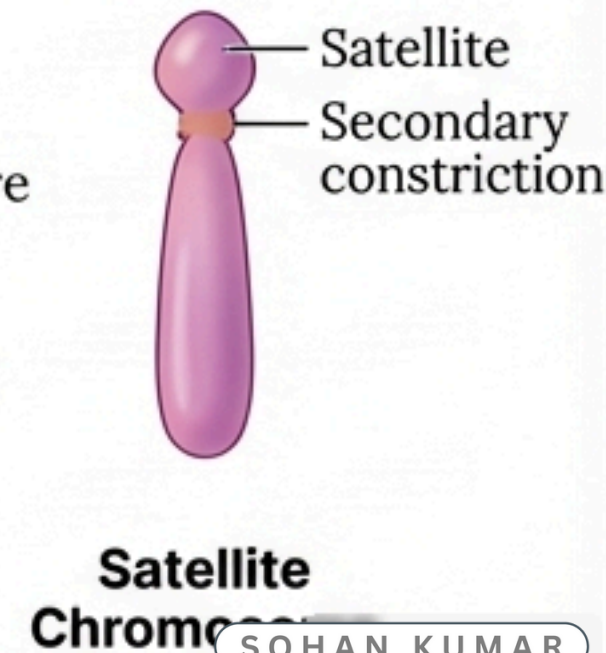
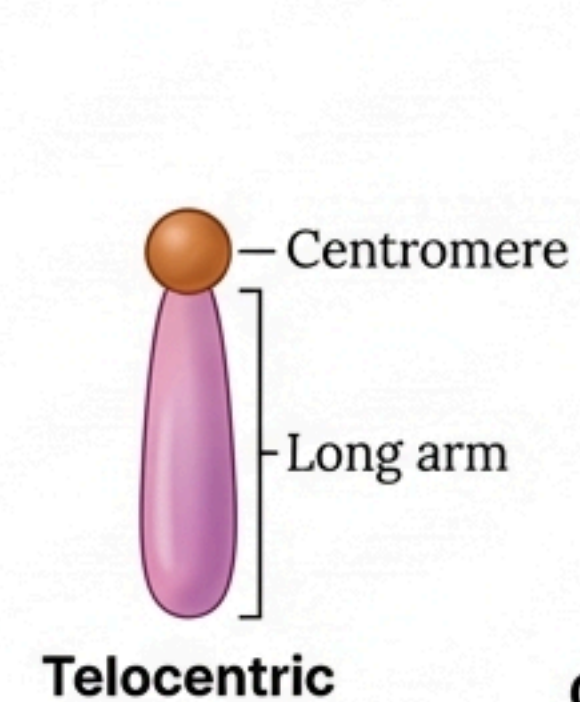
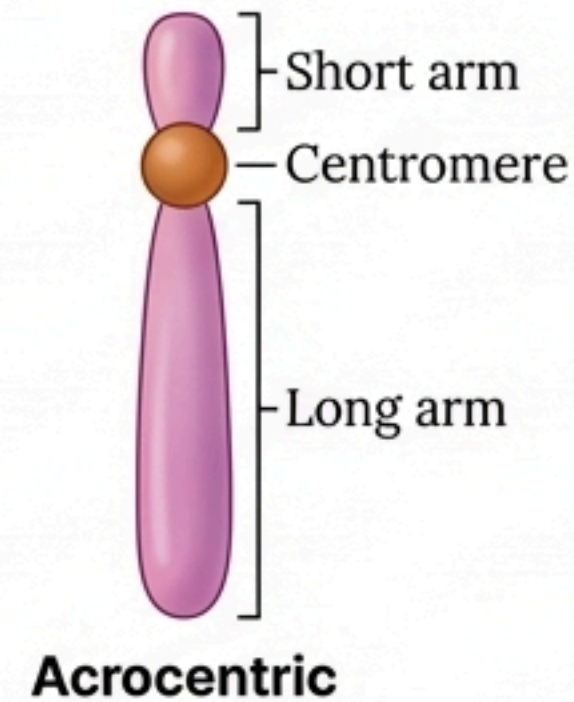
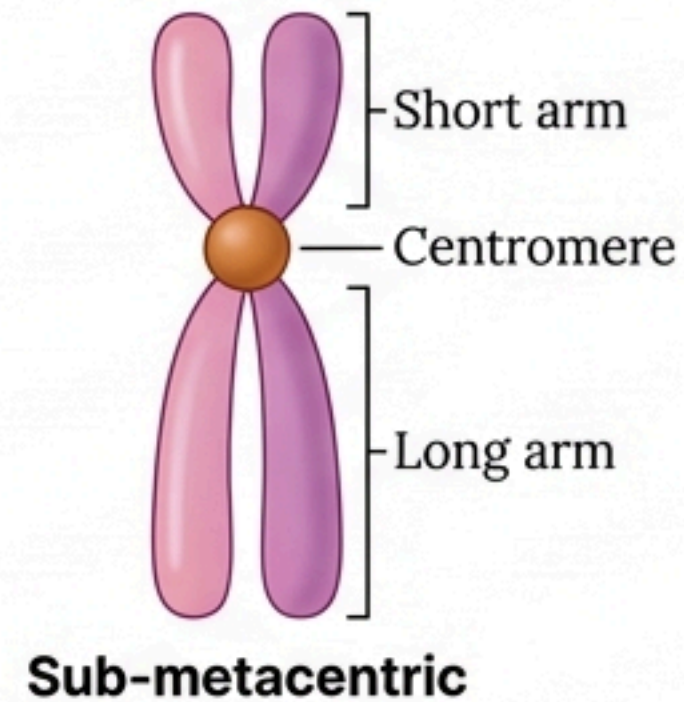
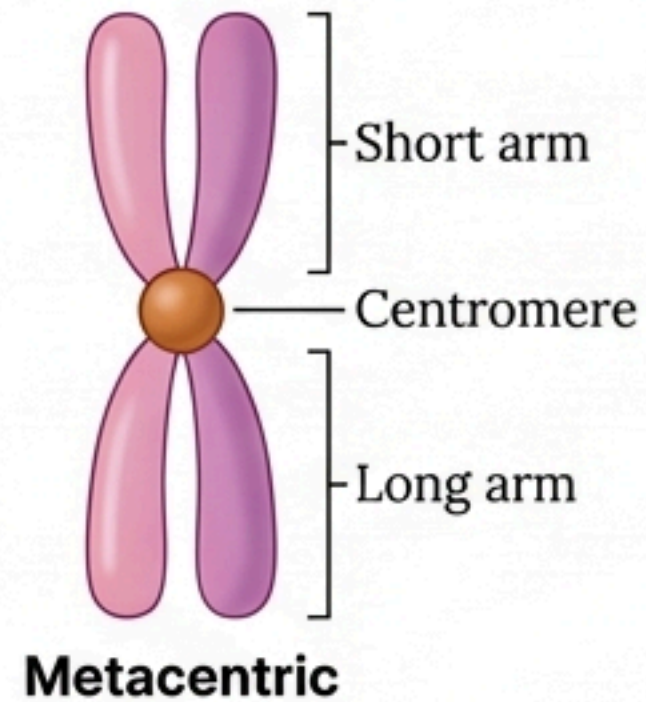
⚠ NEET TRAP: Cilia/Flagella = 9+2 | Centriole = 9+0

The Nucleus & Chromosomes

Nucleus Structure

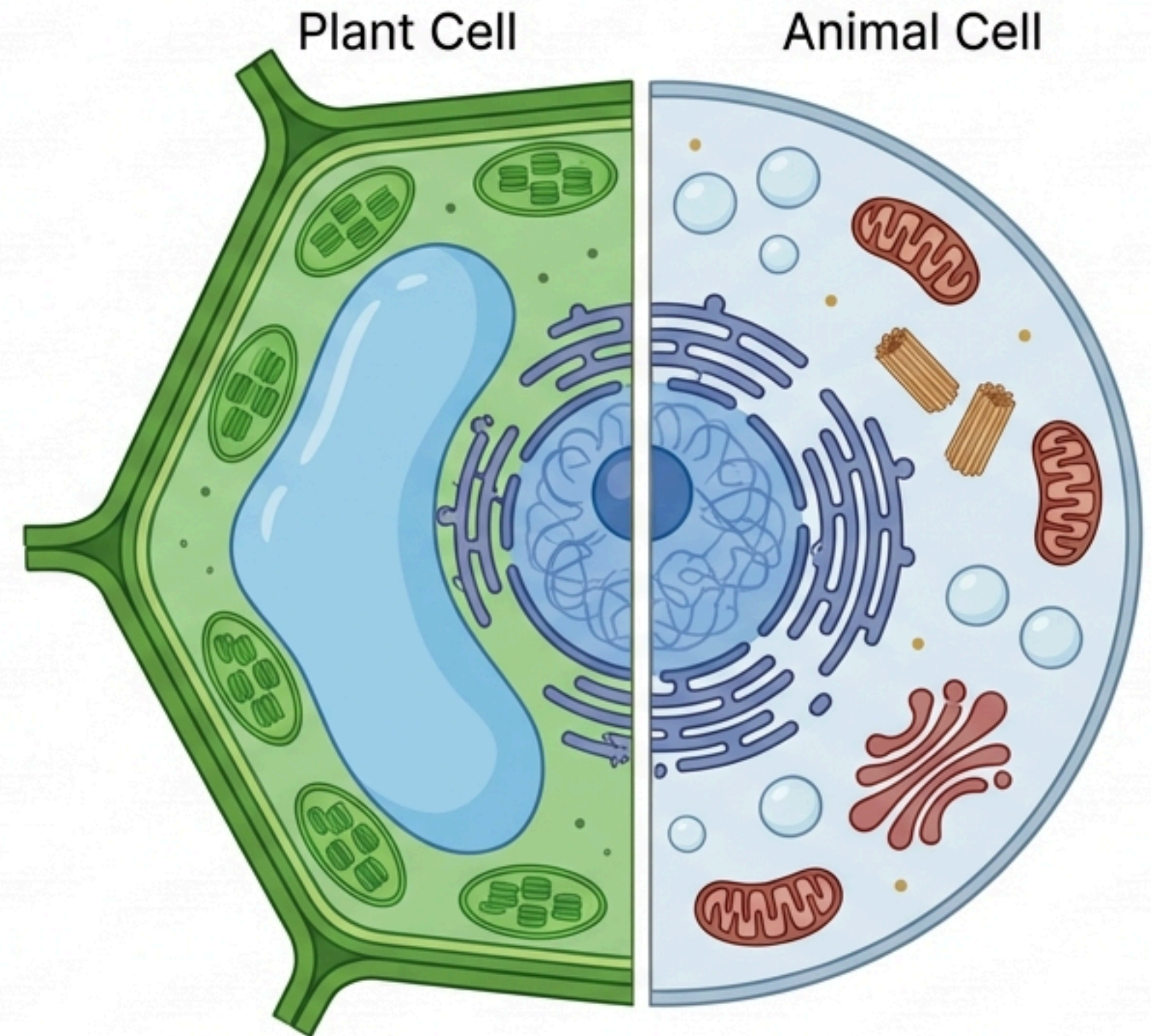


Chromosome Classification (Based on Centromere)



Comparison: Plant vs. Animal Cell

Feature	Plant Cell	Animal Cell
Cell Wall	Present (Cellulose)	Absent
Plastids	Present	Absent
Vacuoles	Large central (90% volume)	Many small / Absent
Centrioles	Absent (mostly)	Present (Cell division)
Nucleus Position	Peripheral (Pushed by vacuole)	Central



NCERT Line – Direct NEET Traps

 **Statement:** Pili and Fimbriae help in bacteria motility.

 **Correction: FALSE.** They are surface structures for attachment/mating only. Flagella = Motility.

 **Statement:** The secondary cell wall grows on the outer side.

 **Correction: FALSE.** It forms on the INNER side (towards the membrane).

 **Statement:** Plasmodesmata connect the cell walls.

 **Correction: FALSE.** They connect the CYTOPLASM of neighboring cells.

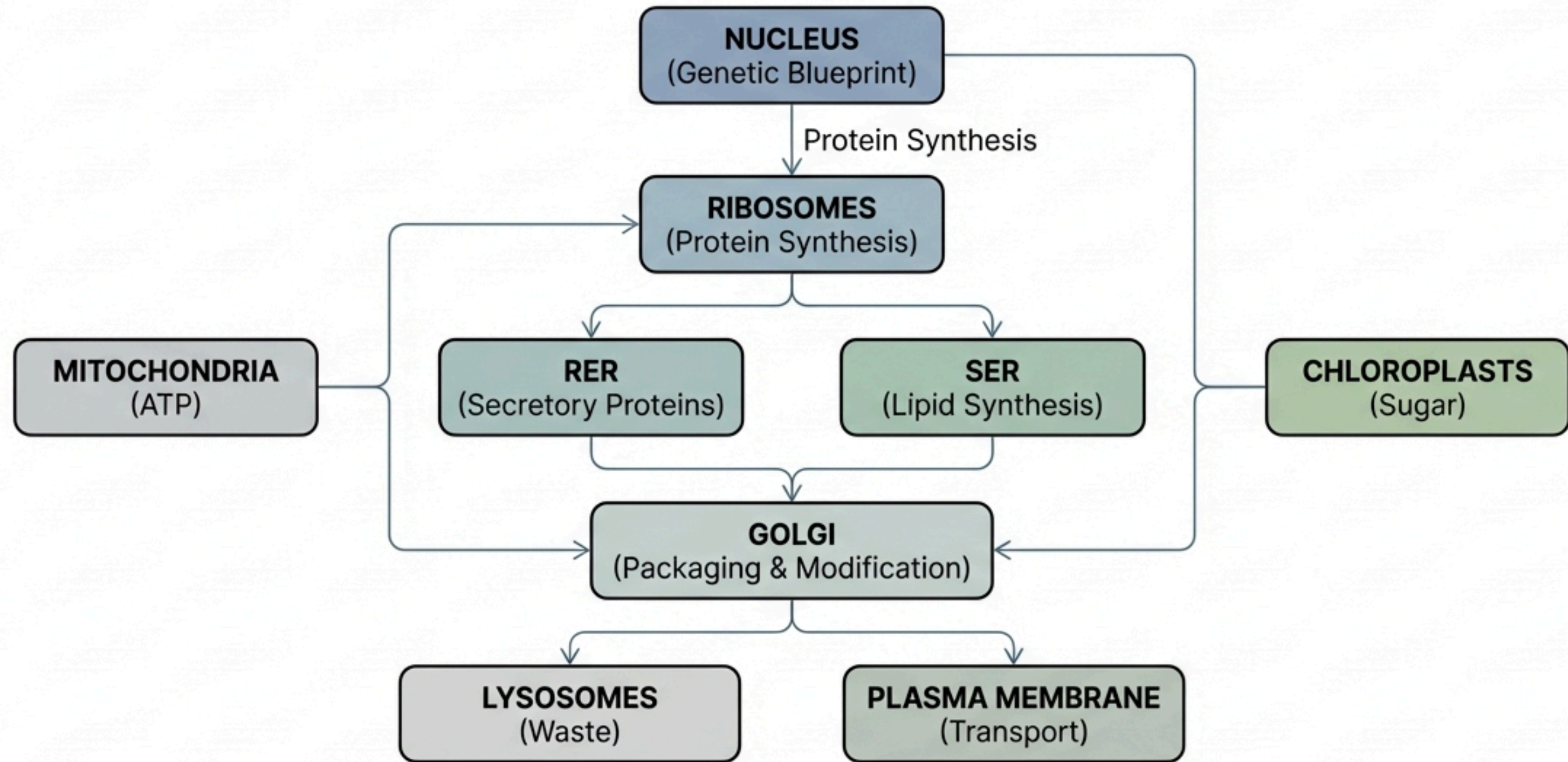
 **Statement:** Ribosomes are membrane-bound organelles.

 **Correction: FALSE.** They are non-membrane bound (in both Prokaryotes & Eukaryotes).

 **Statement:** Steroidal hormones are made in RER.

 **Correction: FALSE.** They are synthesized in the SER (Smooth ER).

Final Chapter Snapshot: Functional Interconnections



Golden Rule: Cell is the structural & functional unit. Anything less than a complete cell ≠ Independent Life.